

Enabling real-time applications in Wi-Fi networks

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Abstract

Being of high importance, real-time applications, such as online gaming, real-time video streaming, virtual reality, and remote-control drone and robots, introduce many challenges to the developers of wireless networks. Such applications pose strict requirements on the delay and packet loss ratio, and it is hardly possible to satisfy them in Wi-Fi networks that use random channel access. The article presents a novel approach to enable real-time communications by exploiting an additional radio. This approach was recently proposed by us in the IEEE 802.11 Working Group and attracted much attention. To evaluate its gain and to study how real-time traffic coexists with the usual one, a mathematical model is designed. The numerical results show that the proposed approach allows decreasing the losses and delays for the real-time traffic by orders of magnitude, while the throughput for the usual traffic is reduced insignificantly in comparison to existing networks.

Keywords

Wi-Fi, real-time applications, time sensitive networks, IEEE 802.11ax, ultra reliable low latency communications, preemptive service, busy tone

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Introduction

Many applications that can be classified as real-time applications (RTAs) have been gaining momentum during the recent years. These applications include Internet video surveillance, Internet gaming applications, and virtual and augmented reality. According to Cisco forecasts, all the mentioned categories will increase their traffic approximately 10-fold from 2017 to 2022.¹ Such a demand for real-time services affects the activities of the telecommunication standardization bodies; for example, the IEEE 802 LAN/MAN Standards Committee considers RTA as a highly important use case for the next generation of Wi-Fi networks. Triggered by our proposal in November 2017,² in July 2018 the IEEE 802.11 Working Group created an RTA Topic Interest Group (TIG), which aims at determining and classification of the most essential RTAs for Wi-Fi networks; at formalizing their requirements in terms of data transmission delay,

packet loss rate (PLR), packet error rate (PER), bandwidth and network density; and at proposing solutions for real-time data transmission.³ Possible RTAs include online gaming, video streaming, virtual reality, and remote control. Their delay requirements vary from 1 ms (force control) to 100 ms (logistics), while the PLR requirements vary from 10^{-9} (emergency systems) to 10^{-2} (machine tools), see Table 1.⁴⁻⁶

Obviously, such heterogeneous requirements need different means to satisfy them. For example, heavy

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Table 1. Requirements for some real-time applications considered in RTA TIG.

Application	Latency, ms	Bandwidth	Packet size, bytes	Robustness
Emergency control	1 ... 3	10 ... 100 kbps	<100	PLR < 10^{-9} ... 10^{-5}
Real-time video	3 ... 10	0.1 ... 20 Gbps	>4000	PLR < 10^{-7}
Online gaming	10	0.1 ... 1.0 Mbps	100 ... 200	<3 consecutive loss, PER < 0.1%
Regulatory control/camera input	10	10 ... 100 kbps	<100	PLR < 10^{-9} ... 10^{-5}
Supervisory	10 ... 100	1 ... 10 kbps	<100	PLR < 10^{-9} ... 10^{-5}

PLR: packet loss rate; PER: packet error rate.

flows require a stable high-rate channel. If the packet flow has a constant bit rate (CBR), the devices can arrange a scheduled transmission using several schedule-based packet transmission mechanisms included in the Wi-Fi standard, for example, Hybrid Coordination Function Controlled Channel Access (HCCA).⁷ When the packet flow is sporadic, that is, the packets arrive at the transmitter unpredictably, a fixed long schedule is inefficient, and such a case is the most challenging for wireless networks developers and is studied in the article.

A similar problem has been carefully studied by 3GPP which aims at enabling Ultra-Reliable Low Latency Communications (URLLC) as an essential feature for the upcoming 5G Cellular Systems. Although similarly to RTA TIG, they consider a wide palette of use cases with different requirements, typically, URLLC means the packet losses less than 10^{-5} and delay less than 1 ms.⁸ Such strict requirements cannot be satisfied without significant changes in the existing network protocols. A good example of changes induced by URLLC is the New Radio air interface for the cellular networks, which significantly reduces the time granularity of transmissions, since the current LTE scheduling granularity of 1 ms (and even greater delay between request for resources and the grant) is too long to satisfy the URLLC delay requirement.^{9,10}

Wi-Fi networks do not suffer from the time granularity problem relevant to LTE, but reliable data delivery is still a challenge since Wi-Fi stations (STAs) operate in unlicensed spectrum and use random channel access for transmission. Naturally, frame collisions cause packet losses and increase delay. Another significant issue is that a Wi-Fi STA should wait for the channel to become idle before starting its transmission; however, transmissions of other STAs can be rather long, the upper bound being approximately 5 ms. Since a legacy Wi-Fi STA has no means to stop an ongoing long frame transmission of another STA, with the current Wi-Fi it is impossible to satisfy the delay requirement of 1 ms.

In the article, we consider the ways (discussed in the literature and RTA TIG) which can be used to reduce the latency in future Wi-Fi networks. The main

contribution of this article is the following. First, we propose a new method that can provide instant channel access for RTA traffic. Second, we develop a mathematical model of the proposed method that provides accurate estimates of the PLR and delay distribution as functions of the network and traffic parameters. Third, we use the developed mathematical model to evaluate the efficiency of the method and to find the maximal load, for which our method satisfies the requirements of RTAs. Finally, we evaluate the impact of the proposed method on non-RTA traffic. Specifically, we show that throughput degradation for non-RTA traffic is rather small.

RTA challenges

Channel access in Wi-Fi networks

In modern Wi-Fi networks, the STAs access the channel to transmit a frame with the Enhanced Distributed Channel Access (EDCA), which works as follows. Every STA adds newly generated frames to its transmission queue. If a frame arrives in an empty queue, the STA senses the channel, and if the channel is free, the STA transmits the frame at once. If the channel is busy, the STA initializes a backoff counter with an integer random variable equiprobably drawn from the interval $[0, W_r - 1]$, where r is the retry counter, initially equal to 0, and W_r is the contention window, calculated as

$$W_r = \begin{cases} W_{min}, & r = 0 \\ \min(2W_{r-1}, W_{max}), & r > 0 \end{cases}$$

Here W_{min} and W_{max} are the minimal and maximal contention window, respectively.

When the channel is busy, the STA suspends its backoff counter. If the channel becomes idle for time $AIFS$, the STA resumes its backoff counter and decrements it every time interval σ . When the counter reaches zero, the STA transmits the frame. The recipient STA must transmit an acknowledgment (ACK), *SIFS* after the arrival of the frame. The sender STA waits for an ACK for an AckTimeout interval, and if the ACK arrives, removes the successful frame from

the queue and sets the retry counter to zero. If no ACK arrives, the frame is considered lost, the STA increases its retry counter, draws a new backoff value, and makes a new transmission attempt.

EDCA distinguishes between four access categories (ACs): voice, video, best-effort, and background. The STA has a separate queue for each AC and a separate backoff function. Moreover, each backoff function has its own parameters W_{min} , W_{max} , and $AIFS$, which equals $SIFS + \sigma \times AIFSN$, where $AIFSN$ is an integer from 2 to 15. By varying these parameters at the STAs, the access point (AP) can prioritize different types of traffic. Thus, a voice packet typically wins the contention with the background one.

Note that EDCA can only provide probabilistic quality of service (QoS) guarantees, that is, the ability to guarantee QoS requirements is subject to the channel load and interference. Even if the AP tries to change the EDCA parameters, there is no guarantee that all the associated STAs follow the recommendation, to say nothing about the STAs associated to the neighboring APs but sharing the same channel. Thus, it is hardly possible to satisfy the RTA requirements in a highly loaded network with aggressive contention. Therefore to support RTA, the Wi-Fi developers need to consider two problems: (1) how to satisfy tight delay and PLR requirements of RTA traffic in a “friendly” environment that supports RTA and (2) how to reach the best possible performance for RTA in an aggressive environment with highly loaded legacy STAs.

IEEE 802.1 TSN

The problem of providing highly reliable real-time transmission has already been considered in fixed networks. For example, for the Ethernet technology, a set of IEEE 802.1 TSN (Time Sensitive Networking) standards¹¹ have been developed. These standards describe general approaches to achieve reliability by frame replication (802.1CB), path control and stream reservations (802.1Qat), while the low latency is achieved by tight synchronization (802.1AS), time-aware scheduling, traffic shaping (802.1Qbv), and frame preemption service. However, the solutions proposed for the Ethernet do not fully solve the problems arising in the wireless networks, since the modern Ethernet networks consist mostly of point-to-point links and do not suffer from the problem of frame collisions during the channel access in such an extent as Wi-Fi networks. Below, we evaluate how different approaches designed in 802.1 TSN (basically for Ethernet networks) can be used in Wi-Fi networks.

RTA traffic identification. To provide reliable real-time service for a subset of transmitting packets, we need first

of all to identify and differentiate them from other types of traffic. This can be done with a Virtual Local Area Network (VLAN) tag field which is added to the header of Layer 2 frames as specified in 802.1Q. This tag assigns the frame to one of eight traffic classes defined in 802.1Q. Finally, these traffic classes are mapped to different EDCA ACs. Thus, real-time traffic can be mapped to one of four existing ACs. Currently, there is a discussion in RTA TIG on introducing a new AC for RTA traffic, which should have highly prioritized channel access.

Admission control. In Wi-Fi networks, channel access may take much time because of high congestion. For emerging RTA traffic, high congestion may violate QoS requirements. To avoid channel overload, in 802.11 networks admission control can be used. Moreover, it can also be used across overlapping networks if they are managed by the same entity. Unfortunately, because of the unlicensed spectrum, it is possible that at least one overlapping Wi-Fi network is unmanaged, which can spoil admission control decisions. However, there are many scenarios where the network can be managed. Moreover, the requirement to support admission control across overlapping networks may be introduced to the recently released for Wi-Fi operation 6 GHz band, where no legacy devices operate.

Time-aware shaping. Introduced in the IEEE 802.1Qbv standard, time-aware shaping allows the scheduler to define the times when packets from each queue are served in order to avoid congestion among them.

The scheduling problem is being solved at the switch, which has to serve frames from multiple flows arriving at the switch in such an order that guarantees them a specific worst-case latency, which can be formulated as an optimization problem.¹²

Time-aware shaping can be extended to an 802.11 network. In this case, it is important not only to eliminate contention among all the queues within a STA but also to avoid congestion for all STAs operating in the same area, including those associated to another AP. For that, the 802.11 standard provides several mechanisms which allow defining network-wide service periods reserved for RTA traffic and synchronize these periods between several APs. For example, HCCA Transmission Opportunity (TXOP) Negotiation can be used. However, HCCA TXOP Negotiation has several drawbacks which shall be eliminated to guarantee fast and reliable schedule distribution among all STAs.

Time-aware shaping is especially fruitful for regular RTA streams, where the scheduler can predict time instants when the next RTA packets come. In contrast, for random, for example, Poisson flows, time-aware shaping may be inefficient since either the RTA packet

needs to wait a lot till the next scheduled time or the scheduler needs to reserve too much time for RTA transmissions.

Frame replication. Transmission of a packet can be delayed or damaged in a Wi-Fi network because of high contention and interference. Both factors are unpredictable and out of control. Thus, to reduce the delay and increase the reliability of RTA packet transmission, the same packet can be replicated and transmitted over several radio interfaces which operate in different channels and thus, can be statistically independent. Such a dual-link capability has been enabled for Ethernet by the 802.1CB standard and could be extended to 802.11. Dual link can be easily used by devices which operate in two channels (2.4 and 5 GHz). Although such a feature does not enable reliable low latency communication, it can be used in conjunction with other methods to improve their performance.

Frame replication can also be done with several wireless technologies. For example, Grigoreva et al.¹³ use a heterogeneous network which exploits LTE, Wi-Fi, and TETRA (terrestrial trunked radio) connectivity to significantly boost the reliability in emergency scenarios. Specifically, they aim at providing the PLR less than 10^{-5} . However, maintaining several network infrastructures is rather expensive, and inter-technology cooperation introduces additional delay. Therefore, it is important to optimize the reliability and latency for each technology.

Another solution to guarantee the reliability in a local area network is the IEC 62439-3 Parallel Redundancy Protocol (PRP).¹⁴ Its idea is quite straightforward: the reliability is achieved by doubling the network infrastructure and using two interfaces on each device, and every packet in the network is sent simultaneously via both interfaces. Initially designed for industrial Ethernet networks, it has been adapted for wireless networks as well,¹⁵ in which case the parallelization is achieved by making two wireless interfaces use non-intersecting frequency bands. Such an approach has proven to significantly decrease the PLR since the packet is lost only when both channels experience failure at the same time. The PRP is built on top of a link layer solution, that is, it is implemented by connecting a redundancy box to the interfaces used in parallel, which doubles the packets and sends each packet to the corresponding interface. On one hand, it is convenient, because such a box can be used with off-the-shelf devices. On the other hand, processing of the packet at the redundancy box introduces an additional delay which grows with the packet length. Another disadvantage of using an independent box in the network is that the interfaces have separate medium access control (MAC) queues and thus cannot track successful

deliveries made by each other, which leads to unnecessary transmissions, increased channel occupancy and the time that the frames spend in the transmitter's queue, and thus leads to packet losses. In general, PRP over Wi-Fi consumes twice as much channel resources than the ordinary Wi-Fi.

This problem has several solutions, one of which is the reactive duplicate avoidance.¹⁶ Its idea is to modify the standard EDCA procedure in such a way that when a Wi-Fi interface on a device successfully delivers a frame, that is, receives an ACK from the recipient, it generates a cross-acknowledgments by which it notifies the second interface that it can drop the corresponding frame from its queue. The performance of this method has been evaluated¹⁷ in a wide set of experiments in realistic industrial environments, and it has been shown that it allows achieving an average latency less than 1 ms. However, the presented experimental results show that the portion of packets that miss the deadlines of 1 ms or 3 ms is still greater than 10^{-4} , which can be inappropriate for some applications.

The reactive duplicate avoidance can be complemented by the proactive duplicate avoidance,¹⁸ which relies on intentionally displacing operations on physical channels. For example, frames on the second Wi-Fi interface can be deferred for some time relative to the transmission on the first interface. Thus, the probability that the frames will be discarded on the secondary interface is increased, but it introduces a tradeoff between latency and channel usage. Experimental results show that the usage of proactive duplicate avoidance decreases the channel resource over usage to 10% in comparison with ordinary Wi-Fi, while the latency is increased by approximately 100 μ s.

Preemption. Defined in 802.1Qbu and 802.3br, preemption is the ability of the sender to immediately stop the transmission of a low-priority frame, when a real-time frame is enqueued. Preemption is especially important if the packets arrive for transmission in random time instances and the delay budget is smaller than or comparable to the duration of the ongoing transmission. Thus, no schedule can be done in advance, and the only way to satisfy the delay requirement is to stop ongoing transmission. If the RTA packet comes to the device which currently transmits, preemption can be implemented very easy. Otherwise, it is a challenging task in Wi-Fi networks, since there is no way in the current Wi-Fi standard to reliably notify another STA being transmitting to stop the transmission. In the article, we propose and evaluate a way how to implement preemption in Wi-Fi.

Proposed preemptive access scheme

To enable preemption in Wi-Fi networks, we propose to add an AC for the RTA traffic and to use an

auxiliary radio. The auxiliary radio works in a narrow band, providing low-rate signaling, but being very robust. It is only used in the service channel, which is separate from the main transmission channel.

There are many ways to implement the auxiliary radio since its only functionality is to transmit and to detect a busy tone. It can be a continuous busy tone, similar to the one proposed for the busy tone multiple access in.¹⁹ Another way to implement such functionality is to use a simplified radio similar to the one considered in the IEEE 802.11ba²⁰ standard. The 802.11ba radio provides control information transmission using the on-off keying modulation over some subcarriers of the traditional 20 MHz Wi-Fi channels. Such a modulation could be used to transmit a synchronization sequence easily discernible by the receiver.

When the RTA queue is empty, the STA processes other queues and transmits packets in the main channel. In parallel, the STA is sensing the service channel using the secondary radio. As long as it is idle, the STA may use the main channel to transmit non-RTA frames according to the legacy EDCA method.

When the RTA queue becomes non-empty, the STA starts sending a busy tone in the service channel. Having received the busy tone, all the STAs that transmit non-RTA frames should free the main channel at once to let the RTA frames be transmitted (see Figure 1). The STA senses the main channel, and if it becomes idle within a short time δ equal to the maximal time needed by another STA to detect the busy tone and stop its frame transmission, then *AIFS* after the channel has become idle it transmits its frame. The value of δ should be rather small in comparison with *AIFS*. Otherwise, when the channel is busy (it might happen if the transmitting STA is legacy and does not support RTA busy tone or if the transmitting STA itself transmits an RTA frame), it contends for channel access according to the EDCA function of the RTA AC.

It is important that the first transmission attempt is made without waiting for backoff. Such behavior is similar to EDCA transmission when a frame arrives in an empty queue. Since such transmission is made asynchronously, it has the minimal delay and a much lower

probability of collision in comparison to a transmission made after the backoff countdown.

In our approach, we also introduce a delay budget D_{max} , for example, 1 ms. If the STA does not succeed to deliver a frame within D_{max} after its queueing, the frame is discarded.

The final remark about the proposed approach is relevant to non-RTA queues. When the service channel is busy, the EDCA function of non-RTA queues should treat the channel as busy, so the corresponding backoff counters should be frozen. If a STA detects a busy tone during its transmission of a non-RTA frame, it should consider such an event as a frame collision.

The state machine for a STA supporting the preemptive access is shown in Figure 2. On the scheme, states are shown as rectangles; their names are written in bold in the top. Transitions between states are shown with arrows, and labels at the beginning of arrows indicate the events that cause the transitions.

Considered scenario and problem statement

The proposed approach reduces the RTA frame delays at the cost of the throughput degradation for non-RTA frames. At the same time, it does not eliminate the contention between the STAs that transmit RTA frames, so it is essential to find the RTA traffic intensity, for which the proposed approach can satisfy the requirements on the RTA traffic delay and PLR.

To evaluate the performance of the proposed approach, we consider a network that consists of M STAs that generate non-RTA traffic (briefly, regular STAs) and N STAs generating RTA traffic (referred to as RTA STAs).

Regular STAs transmit data in the saturated mode, that is, they always have frames for transmission. RTA STAs are non-saturated: after successful transmission of a frame, a new frame arrives at an RTA STA in random time, distributed exponentially with parameter λ . The probability that a new frame arrives at the STA before time t equals

$$A(t) = 1 - e^{-\lambda t} \quad (1)$$

All the STAs are in transmission range of each other, so there are no hidden STAs.

We consider two cases. In the *preemptive access* case, both RTA and regular STAs support the preemptive access scheme described above. In other words, regular STAs stop their own transmissions once the busy tone appears.

In the *legacy access* case, regular STAs do not have auxiliary radio. Thus, they do not sense busy tone, do not stop ongoing transmission and do contend for the channel with the RTA STAs.

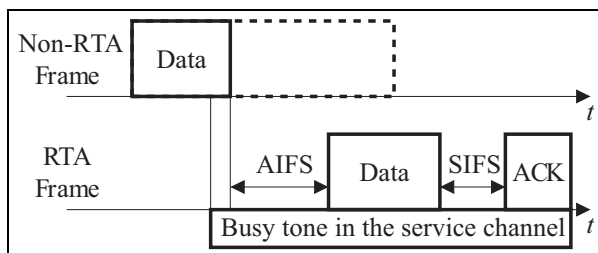


Figure 1. Preemptive access for RTA frames.

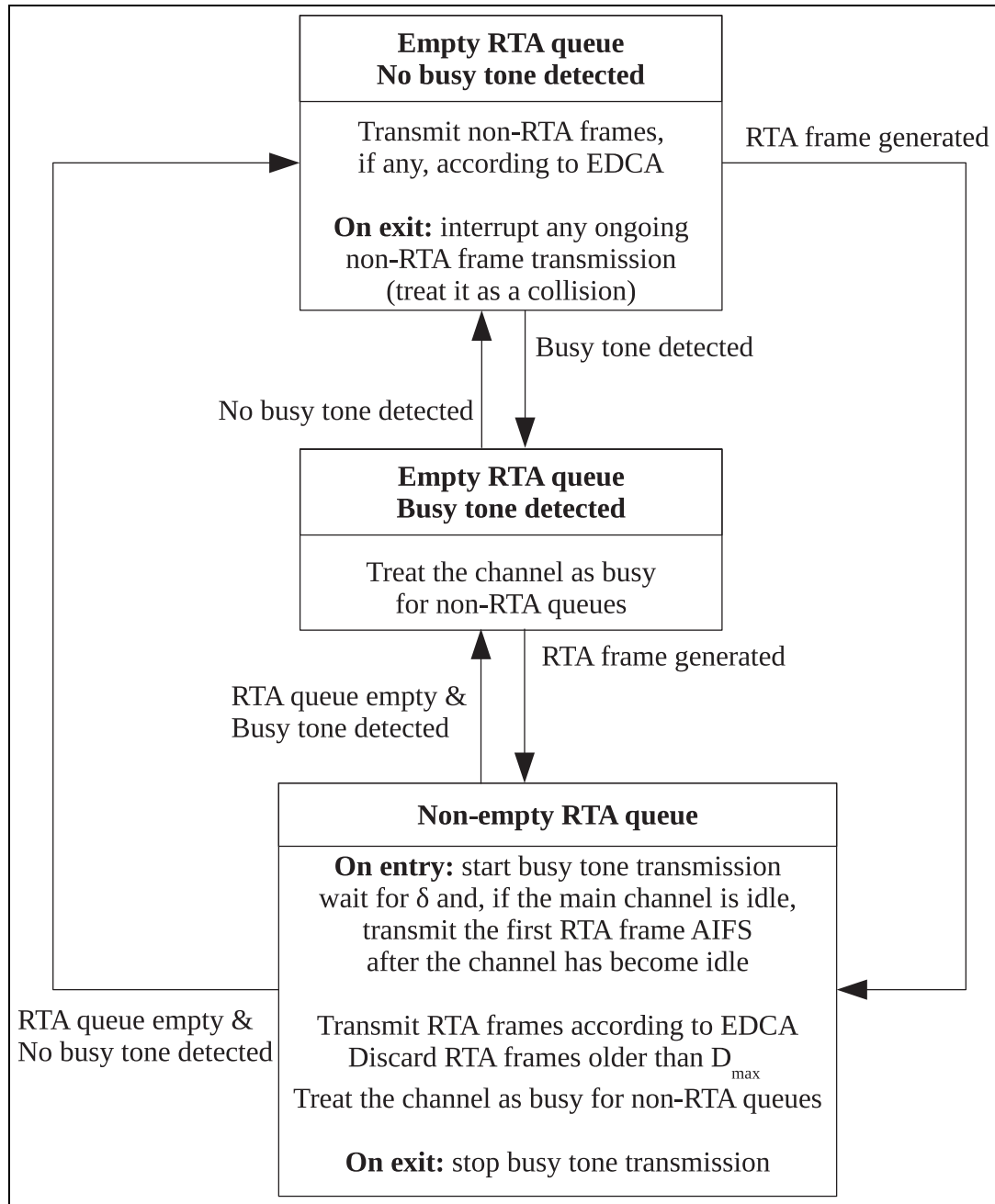


Figure 2. State machine for a STA supporting the preemptive access scheme.

By default, all the analysis and results are presented for the *preemptive access* case, unless additionally specified.

The goal of this article is to evaluate the proposed preemptive access approach in terms of frame delivery delay and PLR for the RTA STAs. Apart from that, we need to find the limits of the proposed approach, that is, such RTA traffic intensity, which makes the scheme viable. An additional goal is to compare the results for the preemptive access scheme with legacy channel access and thus to estimate the possible gain from using the proposed approach.

Analytical model

To evaluate the proposed approach, we develop a mathematical model of the described network. The modeling approach that we use to describe the behavior of the RTA STAs is similar to the study by Bianchi.²¹ The main difference between our model and the one presented by Bianchi²¹ is that Bianchi describes the transmission of saturated data flows, that is, the STAs in the network always have frames for transmissions, while we describe the transmission of non-saturated data flows generated by the RTA STAs. Another

important difference is that in our model RTA frames are dropped if their delay reaches D_{max} , while the model by Bianchi²¹ does not take into account such restrictions. We consider that an RTA STA can be in a state when it has an empty transmission queue and that it can transmit its frame without the backoff countdown if the frame is generated when the channel is idle. Note that, due to the prioritized access of RTA frames, the behavior of RTA STAs is not affected by the regular STAs. Since we develop the model in order to find the PLR of RTA STAs, we consider only the N RTA STAs. According to the scenario, all the STAs can sense each other, which means that all the RTA STAs count their backoff counters simultaneously, so we observe the network at the beginning of the RTA frame transmissions and backoff countdowns.

The interval between the time instants when we observe the network is denoted as a *slot*. Similarly to Bianchi,²¹ there are several types of slots of different duration. There are *empty* slots, in which no STAs try to transmit their frames and which last for σ . There are *successful* slots, in which only one STA tries to transmit its frame which is delivered successfully because we assume that the used modulation and coding scheme (MCS) is efficient enough to cope with noise. Such slots last for $T_s = T_{Data} + SIFS + T_{Ack} + AIFS$, where T_{Data} is the duration of an RTA frame and T_{Ack} is the duration of an acknowledgment. Also, there are *collision* slots, which are chosen by more than one STA. We consider that a situation, when several STAs try to transmit their frames simultaneously, always ends in an unsuccessful transmission attempt. Since in the studied scenario all RTA STAs transmit frames of the same duration T_{data} , the collision slots last for $T_c = T_{Data} + AckTimeout$.

Let us select an RTA STA and describe its operation with a discrete-time Markov chain, the time unit of which is a *slot*. When the STA has no frame for transmission, it is in the *Idle* state. When the STA generates a frame, and the channel is free, it instantly transmits its frame and remains in the *Idle* state. Such a transmission is called the *asynchronous transmission*. If the channel is busy, the STA starts counting down its backoff counter. During the backoff countdown, its state is described with a pair (r, b) , where b is the backoff counter value and r is the retry counter at the considered time.

To derive the probabilities of passing from one state to another, we introduce the probabilities τ_a and τ_s that an RTA STA makes an asynchronous and a synchronous transmission during a slot, respectively. These probabilities are the same for all RTA STAs, since they are equal, and we assume that they do not change with time. We also denote the probability that a STA transmits in a slot as $\tau = \tau_a + \tau_s$.

Let us find the probabilities of an empty, successful, and collision slot, provided that the considered STA

does not transmit. The probability of an empty slot is the probability that none of the remaining $N - 1$ RTA STAs transmit

$$p_e = (1 - \tau)^{N-1} \quad (2)$$

A successful slot is a slot in which only one STA makes a transmission attempt and its probability is

$$p_s = (N - 1)\tau_s(1 - \tau_s)^{N-2} + (N - 1)\tau_a \quad (3)$$

The first summand is the probability that out of $N - 1$ RTA STAs only one makes a synchronous transmission and all the other RTA STAs do not make one. The second summand is the probability that any of the $N - 1$ RTA STAs make an asynchronous transmission. Here we assume that an asynchronous transmission is always successful, so if any RTA STA starts an asynchronous transmission, the other STAs sense that the channel is busy and defer their transmissions.

A collision slot is neither successful nor empty, therefore

$$p_c = 1 - p_e - p_s \quad (4)$$

The STA passes from the *Idle* state to backoff countdown if during a successful or a collision slot it generates a frame. The probability of such an event equals

$$p_b = p_s A(T_s) + p_c A(T_c) \quad (5)$$

In a similar manner, we find the probability of an asynchronous transmission

$$p_{at} = p_e A(\sigma) \quad (6)$$

Let us now describe the synchronous transmission of RTA STAs. Let the STA be in the state $(r = i, b = k)$. When b is greater than 1, the STA just decrements the backoff counter at the slot boundary. When b reaches 0, the STA makes a transmission attempt, which ends up with a collision in all cases except when all the other $N - 1$ RTA STAs do not make a synchronous transmission attempt. The probability of collision is

$$p = 1 - (1 - \tau_s)^{N-1} \quad (7)$$

Then the STA passes to the *Idle* state with probability $1 - p$, or with probability p randomly draws a new backoff from the possible W_{r+1} variants. We also denote m as the maximal number of consecutive transmission attempts that can be done within the delay limit: $m = \left\lfloor \frac{D_{max} - T_s}{T_c} \right\rfloor$. If the STA makes a transmission attempt at the backoff stage $m - 1$, it discards the frame either due to the delay limit, or due to a successful transmission, but in any case it unconditionally transits to the *Idle* state. Hereinafter we assume that m is small and W_{max} is greater than $W_{min}2^m$. If we denote

$Pr(state2|state1)$ as the probability that the STA passes from $state1$ to $state2$, we obtain the following system of transmission probabilities of the Markov chain

$$\begin{cases} Pr(i, k-1|i, k) = 1; 0 < k < W_i, 0 \leq i < m \\ Pr(i, k|i-1, 0) = \frac{p}{W_i}; 0 \leq k < W_i, 0 < i < m \\ Pr(Idle|i, 0) = 1-p; 0 \leq i < m-1 \\ Pr(Idle|m-1, 0) = 1 \\ Pr(0, k|Idle) = \frac{p_b}{W_0}; 0 \leq k < W_0 \\ Pr(Idle|Idle) = 1-p_b \end{cases} \quad (8)$$

Let us find the stationary distribution of the chain. Let q_{Idle} be the stationary probability of the chain to be in the *Idle* state, and $q_{i,k}$ be the stationary probability of the chain to be in the state ($r = i, b = k$). First, we note that for any $i \in [0, m-1]$

$$q_{i,0} = p q_{i-1,0} = p^2 q_{i-2,0} = \dots = p^i q_{0,0} = p_b p^i q_{Idle} \quad (9)$$

which has an explanation that the STA makes a transmission attempt with retry counter i only when it has generated a frame, transmitted to the backoff countdown mode, and has made i unsuccessful transmission attempts. The intermediate state probabilities for all $k \in [0, W_i - 1]$ are found as

$$q_{i,k} = \frac{W_i - k}{W_i} q_{i,0} = \frac{W_i - k}{W_i} p_b p^i q_{Idle} \quad (10)$$

Thus we have expressed all the probabilities through q_{Idle} . To find q_{Idle} , we write down the normalization condition

$$q_{Idle} + \sum_{i=0}^{m-1} \sum_{k=0}^{W_i-1} q_{i,k} = 1 \quad (11)$$

which yields

$$q_{Idle} = \left[1 + \frac{p_b}{2} \left(\sum_{i=0}^{m-1} p^i + W_0 \sum_{i=0}^{m-1} (2p)^i \right) \right]^{-1} \quad (12)$$

and for $p \neq 1$ and $p \neq 0.5$ it can be simplified as

$$q_{Idle} = \left[1 + \frac{p_b}{2} \left(\frac{1-p^m}{1-p} + W_0 \frac{1-(2p)^m}{1-2p} \right) \right]^{-1} \quad (13)$$

We express the transmission probabilities in terms of q_{Idle} as

$$\tau_a = q_{Idle} p_{at} \quad (14)$$

$$\tau_s = \sum_{i=0}^{m-1} q_{i,0} = p_b q_{Idle} \frac{1-p^m}{1-p} \quad (15)$$

$$\tau = \tau_a + \tau_s = \tau_s \left(1 + \frac{(1-p)p_{at}}{(1-p^m)p_b} \right) \quad (16)$$

Equations (2)–(7) and (13)–(16) form a system which can be solved numerically to find all the aforementioned values.

The main purpose of our model is to find the PLR for RTA frames. An RTA frame is lost when it cannot be transmitted within D_{max} , so to find the PLR we derive the delay distribution.

Let the STA generate an RTA frame. The delay equals to its minimal value T_s when the frame arrives at the RTA queue while the channel is idle and the STA makes an asynchronous transmission. The probability of such an event is

$$\nu = \frac{p_e \sigma}{p_e \sigma + p_c T_c + p_s T_s} \quad (17)$$

which is the portion of time when the channel is idle.

With probability $1 - \nu$ the frame is generated while the channel is busy with another STA's transmission. The STA, first, waits until the channel becomes idle. The distribution of the waiting time can be found as

$$F_{wait}(t) = \frac{\Pr(T_s - x < t | x < T_s)}{\Pr(x < T_s)} \quad (18)$$

$$= \begin{cases} 0, & t < 0 \\ \frac{e^{\lambda(t-T_s)} - e^{-\lambda T_s}}{1 - e^{-\lambda T_s}}, & 0 \leq t \leq T_s \\ 1, & t > T_s \end{cases} \quad (19)$$

where x denotes the time from the beginning of the transmitted frame to the instant when the new frame has been generated.

When the channel becomes idle, the STA starts the backoff countdown and the following transmission attempts. Let $d_i(e, s, c)$ be the probability of the STA waiting for e empty, s successful, and c collision slots during the transmission attempt i . It can be expressed as

$$d_i(e, s, c) = \frac{1}{W_i} \frac{(e + s + c)!}{e!s!c!} p_e^e p_s^s p_c^c \quad (20)$$

if $0 \leq e + s + c < W_i - 1$ and 0, otherwise. In this formula, the first multiplier is the probability of the considered STA selecting backoff $e + s + c$, and the rest is derived from the multinomial distribution.

With d_i we can find the probability $w_i(e, s, c)$ of the STA waiting for e , s , and c empty, successful, and collisional slots, respectively, during the transmission attempts from 0 to i . It is found as

$$w_0(e, s, c) = d_0(e, s, c) \quad (21)$$

$$w_i(e, s, c) = \sum_{e_i, s_i, c_i} d_i(e_i, s_i, c_i) \quad (22)$$

$$\times w_{i-1}(e - e_i, s - s_i, c - c_i), \quad i > 0$$

which leads to the delay cumulative distribution function (CDF)

$$F_D(t) = \nu + (1 - \nu) \sum_{i=0}^{m-1} p^i (1 - p) \sum_{e, s, c} w_i(e, s, c) \quad (23)$$

$$\times F_{wait}(t - \sigma e - (s + 1)T_s - (c + i)T_c)$$

if $t \geq T_s$ and 0, otherwise. Here the first summand corresponds to the asynchronous transmission, and the second summand includes all possible transmission attempts which are located within the time t . Inside the second summand, we sum over all possible numbers of retries i , p^i is the probability of i consequent retries, and $(1 - p)$ is the probability that the transmission attempt $i + 1$ is successful. The inner sum gives the probability of all combinations of empty, successful, and collision slots which last less than t . It should be noted that in this sum inside F_{wait} we add 1 to s and i to c in order to take into account the transmissions of the considered STA.

With the CDF of the RTA frame delay, the PLR for the RTA frames is found as $1 - F_D(D_{max})$.

Numerical results

To verify the developed mathematical model and to evaluate the efficiency of the proposed approach, we have carried out extensive simulation of the considered scenario.

We consider a network with an AP, 10 regular STAs that generate non-RTA frames in the saturated mode, and several RTA STAs. Each RTA STA generates a Poisson flow of packets with rate λ . All the devices in the network use the IEEE 802.11ax PHY,²² which operates in a 40-MHz frequency band using MCS 5 with one spatial stream. The RTA STAs transmit 40-byte frames, while regular STAs transmit 15-kilobyte frames, and their frame durations equal $53.6 \mu s$ and $924 \mu s$, respectively. For RTA frames, the delay budget is 1 ms.

Figures 3 and 4 show how PLR and the average delay of delivered frames depend on the number of RTA STAs, and consequently on RTA load. First of all, we emphasize that the values obtained with the mathematical model are very close to the values obtained by simulation, the relative error not exceeding 3%. The discrepancy between the mathematical model and the simulation increases with the number of RTA STAs. With a high number of RTA STAs, and consequently a high load, the probability of a slot being occupied by an RTA transmission increases. So, the probability that the STA has to discard its frame long before reaching the limit on attempts m becomes

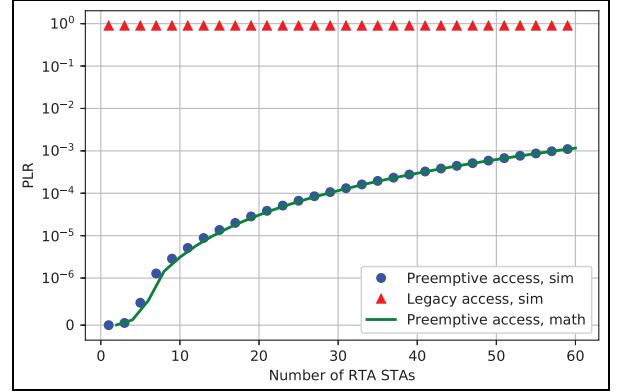


Figure 3. Dependency of PLR for RTA frames on the number N of RTA STAs, $\lambda^{-1} = 50ms$.

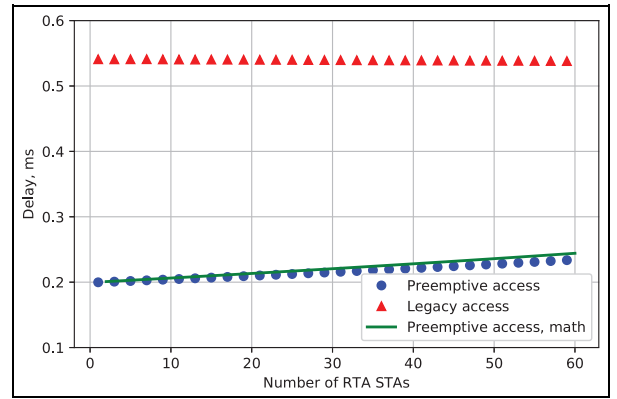


Figure 4. Dependency of average delay for RTA frames on the number N of RTA STAs, $\lambda^{-1} = 50ms$.

notable, while our mathematical model does not consider such a case. At the same time, the number of STAs and λ values, for which the relative error is high, corresponds to the absolute PLR values which are out of the region of interest for the RTA scenarios. Thus, we conclude that the developed mathematical model is accurate enough to evaluate the efficiency of the studied approach.

When the number of RTA STAs is low, the losses in the network are negligible. While the number of RTA STAs increases, their transmissions start to collide more and more often. It leads to additional transmission attempts and increases packet delays. Although the average delay remains well below the required value of 1 ms, the distribution of delays has a heavy tail, and the PLR grows. Note that in the legacy case, almost all RTA frames are discarded because the RTA STAs cannot access the channel within the packet delay budget. At the same time, for those RTA frames which are successfully transmitted within 1-ms delay in the legacy case, the obtained average delay is slightly lower than

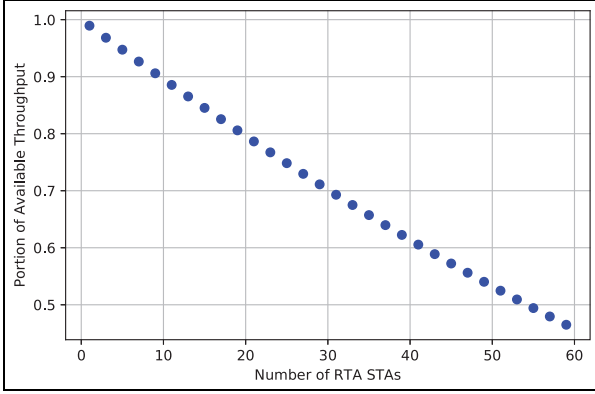


Figure 5. Dependency of the throughput available for non-RTA frames relative to the legacy access on the number N of RTA STAs, $\lambda^{-1} = 50\text{ms}$.

$$\frac{T_{Data}^{Reg} + SIFS + T_{Ack} + AIFS}{2} + \frac{W_{min} - 1}{2}\sigma + T_{Data} \quad (24)$$

where T_{Data}^{Reg} is the duration of a regular data frame transmission. The first summand is half of the regular data frame transmission including the inter-frame intervals and ACK, and the second summand is the average backoff timer made during the first transmission attempt of an RTA STA. Such a result means that in the legacy case, RTA frames are successfully delivered mostly when they are generated during a regular frame transmission, and the RTA STA waits for the channel to become idle, then it waits for its random backoff and after that makes a successful transmission. The RTA frames are also successfully delivered if they are generated while the channel is idle and therefore an asynchronous transmission happens, but such a situation is rather rare since the channel is mostly occupied by the regular STA transmissions. At the same time, the situations when at least two consequent regular frame transmissions happen during an RTA frame transmission lead to the packet losses.

Figure 5 shows that in the preemptive access case, the throughput of regular STAs almost linearly decreases with the load of RTA STAs. Since the load of RTA traffic is low, approximately, each RTA packet reduces the channel time available for regular traffic by half of the average duration of a regular packet plus the duration of the RTA packet. Indeed, RTA packet appears asynchronously (equiprobably during the duration of the regular packet), and stops transmission of the regular packet for the duration of the RTA transmission. After that, the regular packet can be transmitted again from the very beginning.

Figure 6 presents capacity region for RTA traffic in the considered scenario, that is, all points below the curve correspond to the scenario parameters, namely

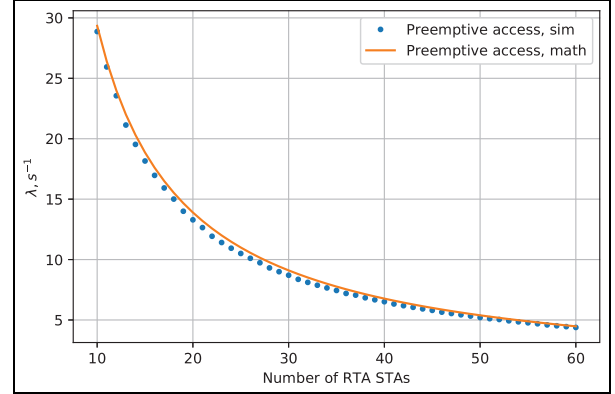


Figure 6. Dependency of the traffic intensity corresponding to $PLR = 10^{-5}$ on the number N of RTA STAs.

the number of RTA STAs and the traffic of each RTA STA, for which the requirements on the delay less than 1 ms and PLR less than 10^{-5} can be met. The obtained results show that the capacity bound is high enough for the typical Wi-Fi scenarios.

Conclusion

RTAs present a significant challenge for the new generation of Wi-Fi networks since they impose strict requirements on latency and reliability. Modern Wi-Fi networks, with their current architecture and channel access procedure, can barely satisfy these requirements, so new approaches to the new traffic service need to be devised.

An essential issue for real-time traffic is how to provide immediate channel access. In this article, we have proposed a preemptive service approach for RTAs. This approach is based on the usage of two radio interfaces, the primary interface being the same as in traditional Wi-Fi networks, and the secondary interface being a narrow band, low rate but very robust. The primary interface is used for data transmission, while the secondary interface is used only to signalize about the presence of the RTA traffic.

We have developed a mathematical model of the proposed approach, which shows that the approach can provide data delivery within 1 ms with PLR less than 10^{-5} for a realistic number of RTA STAs connected to a single AP. The obtained results confirm that this approach is viable and can be used for RTAs. With the model, we also have found the bounds on the number of STAs and the traffic intensity, for which the approach guarantees the satisfactory service. The results show that the proposed scheme can be considered as a candidate for the future RTA standard.

As a future direction of research, we plan to consider a combination of the preemptive access scheme with the OFDMA (orthogonal frequency-division multiple

access)-based channel access, introduced in the Wi-Fi networks with the new IEEE 802.11ax amendment, as a solution to provide real-time service to numerous applications.

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